

Isaac Nathan da Silva Barbosa

FULL-STACK & MOBILE SOFTWARE ENGINEER · AI · APPLIED SECURITY — COMPUTER ENGINEER (5TH SEMESTER) — BUILDING END-TO-END PRODUCTS

PWD — ASD (ICD-11: 6A02.2) + ADHD (ICD-11: 6A05.2)

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SUMMARY

Python · FastAPI · React 19 · Azure Cosmos DB · GPT-4.1 · Flutter · PyTorch — 10 months of professional experience. From Industrial Chemistry to Computer Engineering: built ForestAI from scratch (Stack Overflow + Thonny, no AI-assisted coding). At Paware, migrated hundreds of GB to Azure Cosmos DB (Meritage Homes, USA) and architected LLM-orchestrated pipelines for HelloSocial (GPT-4.1, DALL-E 3, Flux). I learn by solving real problems — from cross-platform MIME types to ReAct agents with schema enforcement.

PROFESSIONAL EXPERIENCE

Full-Stack & Mobile Software Engineer · Freelance Inovesa Florestal

May 2026 — Present

- Lead developer and architect of end-to-end solutions for forest inventory and logistics operations.
- Built and deployed Urutau — Flutter offline-first app with custom sync engine, operating in no-coverage areas.
- Developed Orca — operational planning engine (Python/FastAPI) for large-scale forest restoration, with automatic executive dossier generation, team allocation and optimized schedules.
- React/TS admin panels with auth, reports and multi-format export (XLSX, PDF, CSV).

Software Engineer Meritage Homes

Nov 2025 — Dec 2025 · 2 months · Remote

- Migrated 500GB of critical data to Azure Cosmos DB with zero downtime, ensuring high availability and system resilience.
- Architected AI-driven creative pipelines leveraging GPT-4.1, DALL-E 3, and Flux to automate creative asset generation.
- Developed cross-platform chatbot infrastructure optimized for iOS and Android, featuring context-aware intent routing.
- Engineered PDF parsing pipelines optimized for LLM ingestion, enabling accurate translation and transcription into Spanish and Portuguese.
- Conducted E2E probing and UAT simulating realtors' workflows to optimize UX and platform stability.

Full-Stack Developer & AI Engineer

Oct 2025 – May 2026

Paware Softwares

- Migrated +500 GB legacy data (Meritage Homes, USA) to Azure Cosmos DB — automated extraction pipeline, strict schema validation and auto rollback, zero downtime.
- Resolved cross-platform MIME type inconsistencies (Android native vs iOS octet-stream), creating abstraction layer that eliminated Cosmos DB ingestion failures.
- Architected async media generation pipeline (HelloSocial) — orchestration of GPT-4.1, Flux Kontext Pro, DALL-E 3 and design APIs (Canva/Placid) with local SQLite + PIL fallback.

Independent Security Researcher

Pesquisador de Segurança Independente · Responsible Disclosure

Mar 2026 – Present

- 20+ vulnerabilities reported in Brazilian government/sectoral infrastructure — 5 fixes confirmed via CERT.br/CTIR Gov. Three disclosure waves (Mar–Jun 2026).
- Microsoft/OneDrive phishing infrastructure takedown. Reverse engineering pipeline: custom Base91 decoding (3 alphabets, 176 strings) → IOC extraction → CERT.br/Cloudflare report.

Robotics Instructor

Sep 2025 – Oct 2025

SuperNerds

- Mastered Arduino and LEGO in <2 weeks. Robotics and programming classes for children/adolescents, balancing with night work.

ENGINEERING PROJECTS

Urutau github.com/xAngryBadger/urutau-app [<https://github.com/xAngryBadger/urutau-app>]

- Flutter offline-first app for forest inventory (Urutau) with custom sync service — cascading UUID remapping across related tables and transactional rollback on failure.
- Architecture and sync logic built from scratch; code generated with LLM assistance (web) and manually reviewed. React admin panel with auth, photos, reports, and XLSX/PDF/CSV export.
- Exponential backoff (2s, 4s, 8s), auth retry wrapper with transparent 401 refresh, ngrok bypass for development.

Flutter · Dart · Drift/SQLite · React · PocketBase · TypeScript

Orca github.com/xAngryBadger/orca [<https://github.com/xAngryBadger/orca>]

- Operational planning engine for large-scale forest restoration — automatic executive dossiers, optimized schedules, tariff/crew/territory management.
- FastAPI + Jinja2 + Rich CLI with unit tests. Executive spreadsheet generation (openpyxl) and professional PDFs. Input data validation pipeline with pandas.

Python · FastAPI · pandas · openpyxl · Jinja2 · Rich

HarpIA github.com/xAngryBadger/harpIA [<https://github.com/xAngryBadger/harpIA>]

- Creative automation engine with 9+ AI models (GPT-4.1, Flux, DALL-E 3, Sora, Veo, Nano Banana) — multi-model pipeline with GPT-4.1: copywriting, image search, design compositing, video generation.

- Lightweight stack by default (SQLite + local PIL) with fallback to paid APIs. Backend swap: local SQLite to Azure Cosmos DB + Blob Storage, toggled via env var. Cron-ready with file-locking and stuck batch recovery.

Python · GPT-4.1 · Flux · DALL-E 3 · Sora · Veo · Azure Cosmos DB · SQLite · PIL

ForestAI github.com/xAngryBadger/ForestAi [https://github.com/xAngryBadger/ForestAi]

- Forest species detection and classification with Deep Learning — built from scratch without AI-assisted coding. Stack Overflow + Thonny IDE only.
- Manual annotation of drone images from Fundação Renova (bounding boxes), DeepForest training on local GPU, stratified splits. TensorBoard curve interpretation — detecting memorization vs generalization. The hard way built the intuition that made every subsequent framework click faster.

PyTorch · DeepForest · OpenCV · scikit-learn · TensorBoard

Security Disclosures & Threat Intel github.com/xAngryBadger/security-disclosures
[https://github.com/xAngryBadger/security-disclosures]

- 20+ vulnerabilities in Brazilian government/sector infrastructure — 5 fixes confirmed via CERT.br/CTIR Gov (Mar–Jun 2026). Three responsible disclosure waves: zero retention, multi-channel notification, post-fix verification.
- Microsoft/OneDrive phishing takedown: origin server (procorereviews.com) toppled via controlled flood (~10k requests, sustained HTTP 521). Rotated infrastructure mapped, IOCs extracted, reported to CERT.br/Cloudflare.
- Custom Threat Intel pipeline: Browser → JS download → Custom Base91 decoder (3 unique alphabets, 176 strings) → IOC extraction (PageConfig, tokens, OneDrive redirect) → CERT.br/Cloudflare report.

OSINT · Base91 Decoder · NIST CSF · CERT.br / CTIR Gov · LGPD · Cloudflare · AbuseIPDB · URLhaus

EDUCATION

Computer Engineering
Cruzeiro do Sul

2024 – 2029

5TH SEMESTER (IN PROGRESS)

Chemical Engineering
UFSJ – Campus Alto Paraopeba

2022 – 2024

TRANSITION TO COMPUTER ENGINEERING

CORE STACK

BACKEND & DATA Python · FastAPI · Node.js · SQL · pandas · openpyxl

Python/FastAPI powering Orca planning engine, HarpIA pipelines, and Paware's Cosmos DB migration. pandas/openpyxl for executive dossier generation in Orca.

FRONTEND & MOBILE React 19 · TypeScript · Flutter/Dart · Vite · Tailwind CSS

React 19 + TypeScript → HarpIA frontend, Urutau admin panel, Inovesa institutional site, this portfolio. Flutter/Dart for offline-first Urutau mobile app.

IA & ML PyTorch · DeepForest · Azure OpenAI · Ollama · LLM Agents (ReAct)

HarpIA orchestrates 9+ models (GPT-4.1, Flux, DALL-E 3, Sora, Veo). ForestAI uses PyTorch + DeepForest for species detection. Fennec Excel runs ReAct agents locally via Ollama.

CLOUD & INFRA Azure Cosmos DB · Docker · Terraform · AWS (ECS) · Git · Linux

Migrated 500+ GB to Azure Cosmos DB for Meritage Homes (USA) with zero downtime. Docker containers for AI agents. Terraform for infra-as-code. Custom CachyOS/Hyprrland Linux rice.

SEGURANÇA OSINT · Eng. Reversa (JS/Base91) · NIST CSF · OWASP · Threat Intel

20+ vulnerabilities reported to CERT.br/CTIR Gov (5 fixes confirmed). Custom Base91 decoder (3 alphabets, 176 strings) for phishing JS deobfuscation. NIST CSF methodology for responsible disclosure.

SYNC & OFFLINE Drift/SQLite · UUID remapping · rollback transacional · PocketBase

Urutau offline-first sync: cascading UUID remapping across related tables, transactional rollback on failure, exponential backoff (2s/4s/8s). PocketBase backend with auth retry wrapper.

CERTIFICATIONS

- **Python Essentials 1 & 2** — Cisco Networking Academy
- **Data Science Essentials (pandas, Matplotlib)** — Cisco Networking Academy
- **Data Analytics Essentials** — Cisco Networking Academy
- **Networking Basics (120h)** — Cisco Networking Academy (120h)
- **Introdução à Cibersegurança** — Cisco Networking Academy
- **Segurança em TI** — Fundação Bradesco
- **IA para Otimização de Processos (71h)** — Escola Virtual Gov / Enap / Serpro (71h)
- **Inglês Fluente (3 anos)** — KUMON

LANGUAGES

- Portuguese — Native
- English — Fluent (KUMON, 3 years)